

INSULATION POWER FACTOR REPORT



Equipment Name	2GTG-XF-1.30.EN	Site	SITE3
Serial No.	57005	Power Rating	45000 kVA
Manufacture	ABB	Voltage Rating	118 /11.5 /11.5 kV
Fluid Type	Mineral Oil	Service Type	GSUT

Test Information

Test Instrument	TESTRANO 600
Serial Number	DA212X
Calibration Date	19-Apr-2019

Environment

Ambient Temp.	34 °C
Oil Temp.	36 °C
Winding Temp.	40 °C
Humidity.	68 %

Primary Winding Test

No of Winding : 3

Winding Capacitance Factory/Previous Test

		Voltage Rated		3-5 kV	> 5-12 kV	> 12 kV
		AC Test Voltage		2.5 kV	5 kV	10 kV
Measurent	Capacitance (pF)	Measurent	Capacitance (pF)	Measurent	Capacitance (pF)	
CH+CHL	32209.28	CL+CLT		CT+CHT	38226.63	
CH	13296.52	CL(T)		CT	19112.82	
CHL	18993.76	CLT		CHT	19113.81	

Test Voltage 1 kV

Note: Test included plugin HV power cable (GIS)

Test Connection				Current (mA)	Power (mW)	Power Factor		Capacitance		Measured
Energize	GST	GSTg	UST			Meas. (%)	at 20°C (%)	Meas. (pF)	Error (%)	
HV	LV	-	-	142.33	4268.8	0.2999	0.210	45300.9	40.65	CH+CHL
HV	-	LV	-	126.72	3978.82	0.314	0.220	40333	203.34	CH
HV	-	-	LV	15.6	280.76	0.18	0.126	4965.7	73.86	CHL

Secondary Winding Test

Test Voltage 5 kV

Test Connection				Current (mA)	Power (mW)	Power Factor		Capacitance		Measured
Energize	GST	GSTg	UST			Meas. (%)	at 20°C (%)	Meas. (pF)	Error (%)	
LV	TV	HV	-	31.32	827.96	0.2644	0.185	9967.2	-	CL+CLT
LV	-	HV&TV	-	29.97	775.8	0.2589	0.181	9535.9	-	CL
LV	HV	-	TV	1.35	56.23	0.4165	0.292	431.2	-	CLT

Tertiary Winding Test

Test Voltage 5 kV

Test Connection				Current (mA)	Power (mW)	Power Factor		Capacitance		Measured
Energize	GST	GSTg	UST			Meas. (%)	at 20°C (%)	Meas. (pF)	Error (%)	
TV	HV	LV	-	32.56	971.27	0.2983	0.209	10360.8	72.90	CT+CHT
TV	-	HV&LV	-	31.2	915.64	0.2935	0.205	9929.3	48.05	CT
TV	LV	-	HV	1.35	56.21	0.4164	0.291	431.2	97.74	CHT

Criteria: IEEE C57.152-2013

Type	% Power Factor at 20 °C		
Factory Accetance (FAT)	< 0.5 %		
Service aged	Good	Acceptable	Investigate
Mineral Oil (< 230 kV)	< 0.5%	0.5-1.0%	> 1.0%
Mineral Oil (≥ 230 kV)	< 0.4%	0.4-1.0%	> 1.0%
Natural Ester (ALL)	< 1.0%	-	> 1.0%
Dry Type	< 2.0%	2.0-5.0%	> 5.0%

Capacitance	
% Error	Interpreted
≤ 5%	Normal
5-10%	Questionable
> 10%	Should not be returned to sevice

Response	Tested By	Reviewed By	Approved By
Company	TIS		GLOW
Name	Mr.Chaichaleom W.		Mr.Jirasak S.
Signature			
Date	11-Jun-2019	11-Jun-2019	11-Jun-2019